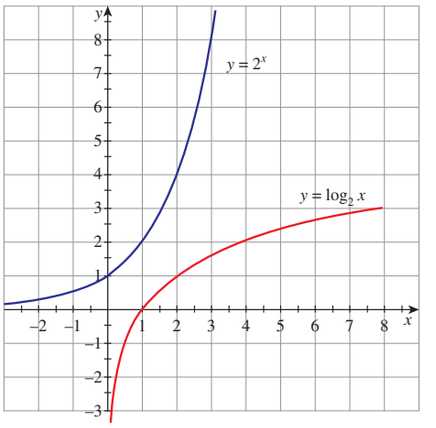


Figure 2.9

Example 2.34 Find the logarithm of 1728 to the base $2\sqrt{3}$.

Solution:
Let $\log_{2\sqrt{3}} 1728 = x$.
Then we have $(2\sqrt{3})^x = 1728 = 2^6\ 3^3 = 2^6\ (\sqrt{3})^6$.
Hence, $(2\sqrt{3})^x = (2\sqrt{3})^6$.
Therefore $x = 6$. That is, $\log_{2\sqrt{3}} 1728 = 6$.

Example 2.35 If the logarithm of 324 to base a is 4, then find a.

Solution:
We are given $\log_a 324 = 4$, which gives
$a^4 = 324 = 3^4\ (\sqrt{2})^4$. Therefore $a = 3\sqrt{2}$.

Example 2.36 Prove $\log \frac{75}{16} - 2\log \frac{5}{9} + \log \frac{32}{243} = \log 2$.

Solution:
Using the properties of logarithm, we have
$\log \frac{75}{16} - 2\log \frac{5}{9} + \log \frac{32}{243} = \log 75 - \log 16 - 2\log 5 + 2\log 9 + \log 32 - \log 243$. (By Quotient rule)
$= \log 3 + \log 25 - \log 16 - \log 25 + \log 81 + \log 16 + \log 2 - \log 81 - \log 3$
$= \log 2$.

Example 2.37 If $\log_2 x + \log_4 x + \log_{16} x = \frac{7}{2}$, find the value of x.

Solution:
Note that $x > 0$.
$\log_2 x + \log_4 x + \log_{16} x = \frac{7}{2}$ becomes $\frac{1}{\log_x 2} + \frac{1}{\log_x 4} + \frac{1}{\log_x 16} = \frac{7}{2}$. (change of base rule)
Thus $\frac{1}{a} + \frac{1}{2a} + \frac{1}{4a} = \frac{7}{2}$ where $a = \log_x 2$. That is $\frac{7}{4a} = \frac{7}{2}$.